System boundary

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System: Inspection Report Routing System

Inside the boundary

Receive Report accepts the Field Team's inspection report and writes the submitted report to D1 Report Register. Check Required Fields checks all eight required fields against W06-S02, sends a validation note to the Field Team when a report is rejected, and passes accepted reports to Resolve Destination. Resolve Destination reads D2 Routing Table to identify the destination. Send Work Item sends the work item to the resolved department and writes a delivery record to D1 Report Register. (W06-S01)

Outside the boundary

The Field Team submits inspection reports and receives validation notes. Electrical, Drainage, and Structures receive routed work items. A service administrator maintains D2 Routing Table outside the modeled boundary; table maintenance is not part of this case. (W06-S01)

Boundary flows

One row per information flow. Repeat an entity if it has more than one flow.

External entity	Direction (into / out of system)	Data flow name	Source
Field Team	Into system	Inspection report (eight required fields)	W06-S01, W06-S02
Field Team	Out of system	Validation note for rejected report	W06-S01
Electrical	Out of system	STREETLIGHT work item	W06-S01, W06-S03
Drainage	Out of system	DRAIN work item	W06-S01, W06-S03
Structures	Out of system	BRIDGE work item	W06-S01, W06-S03

Information movement

Processes: 1 Receive Report; 2 Check Required Fields; 3 Resolve Destination; 4 Send Work Item. Stores: D1 Report Register; D2 Routing Table. Use the flow rows to connect these and the external teams from page 1. Eleven flow rows are provided.

From (entity / process / store)	To (entity / process / store)	Data flow name	Source
Field Team	1 Receive Report	Inspection report	W06-S01
1 Receive Report	D1 Report Register	Submitted report	W06-S01
1 Receive Report	2 Check Required Fields	Received report	W06-S01
2 Check Required Fields	Field Team	Validation note (rejected report)	W06-S01
2 Check Required Fields	3 Resolve Destination	Accepted report	W06-S01
3 Resolve Destination	D2 Routing Table	Asset type / route lookup	W06-S01
D2 Routing Table	3 Resolve Destination	Resolved department destination	W06-S01
3 Resolve Destination	4 Send Work Item	Accepted report + resolved destination	W06-S01
4 Send Work Item	Electrical / Drainage / Structures	Routed work item	W06-S01, W06-S03
4 Send Work Item	D1 Report Register	Delivery record	W06-S01
D2 Routing Table (external maintenance)	Service administrator	Table maintenance is outside system scope; not an internal flow	W06-S01

Balance check

The Field Team's inspection report is the system's external input and appears as the input boundary flow on page 1. The validation note returns to the Field Team as an output. Accepted reports leave through Send Work Item as routed work items to Electrical, Drainage, or Structures. These three department outputs match the three external output flows listed on page 1. The delivery record is stored in D1 Report Register and is not an external boundary flow. D2 maintenance remains outside the modeled system.

Data definitions and finding

Data dictionary — eight required fields

Field	Meaning	Example	Allowed value / format	Source
report_id	Unique reference identifying one inspection report.	CJ-801	CJ- followed by three digits.	W06-S02
asset_id	Reference identifying the inspected asset.	SL-101	Two-letter prefix followed by three digits.	W06-S02
asset_type	Category of the inspected asset.	STREETLIGHT	STREETLIGHT, DRAIN, or BRIDGE.	W06-S02
finding_code	Code describing the inspection finding.	LAMP_OUT	LAMP_OUT, BLOCKED, or SURFACE_DAMAGE.	W06-S02
priority	Indicates the urgency of the finding.	ROUTINE	ROUTINE or URGENT; does not change department.	W06-S02, W06-S03
captured_at	UTC date and time when the report was captured.	2026-10-14T09:00:00Z	YYYY-MM-DDTHH:MM:SSZ.	W06-S02
photo_ref	Fictional reference to an associated photo attachment.	IMG-601	IMG- followed by three digits; no image files supplied.	W06-S02
survey_area	Fictional survey-area label for the report.	NORTH	NORTH, CENTRAL, or SOUTH; not coordinates.	W06-S02

Supported finding

CJ-803, CJ-804, and CJ-807 are BRIDGE reports. The approved destination for BRIDGE reports is Structures (W06-S03), but the delivery log records Electrical as the receiver for all three (W06-S04). This is a routing discrepancy in the recorded outcomes; the evidence does not establish its technical cause.

Unanswered question:

What entries were present in the running D2 Routing Table when these reports were processed, and had any relevant configuration changes or deployments occurred? The running table snapshot and relevant change/deployment history are needed before identifying a cause or claiming a repair.